LexGraph: Proof-Carrying Legal Reasoning from Facts to Authorities, Elements and Outcomes

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Abstract

Language models cite cases that do not exist, and outcome predictors can score well simply by reading the verdict. LexGraph takes only the *facts* of a dispute, as a lawyer holds them before any judgment exists, and returns (i) the statutes and precedents that apply, (ii) element-level support and refutation, each backed by a verbatim quote, and (iii) an outcome estimate with a trace. Every authority and every quotation is checked deterministically against the corpus before it is shown. We work on all 26,688 Indian Supreme Court judgments (1950–2025), concentrating on 6,954 land and property disputes. Preliminary held-out results show that facts alone retrieve precedents at 7.4× a popularity control and statutes at 1.9× the strongest baseline; that the input construction used by published outcome-prediction work inflates AUROC by +0.11 to +0.18 through residual disposition language; and that projecting facts onto an authored element ontology loses outcome signal that a data-induced basis keeps. The remaining work is gold-standard evaluation, signed-graph authority ranking, graph pattern mining and a verifiable demo.

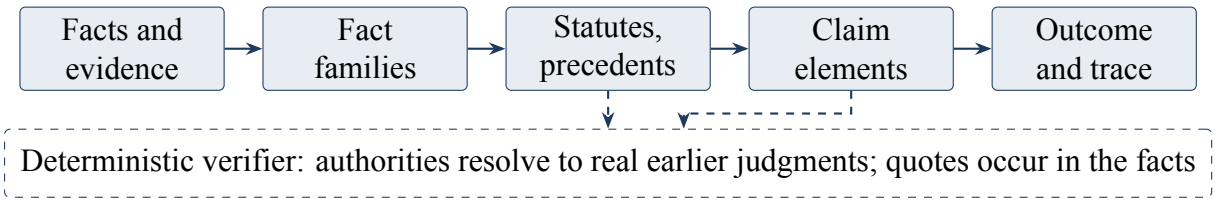
Keywords: legal AI, proof-carrying reasoning, signed citation graph, ontology-directed extraction, precedent retrieval, leakage, Indian Supreme Court.

1. Problem Definition

In practice a lawyer holds facts and evidence, never a decided judgment. The task is therefore:

> **Given the facts of a land or property dispute:** Step 1, predict the statutes and precedents that govern it; Step 2, map facts and authorities to the elements of each claim and say which facts satisfy or defeat which element; Step 3, estimate the outcome, with every output traceable to the record.

Following the course description of a proof-carrying system over a signed proof graph, we make two commitments that separate it from a fluent but unverifiable answer: **no authority is shown unless it resolves to a real, earlier judgment in the corpus**, and **no "satisfied" verdict is accepted unless its supporting quote occurs verbatim in the facts shown**.



We pursue four research questions.

RQ1. How much of the apparent predictability of court outcomes comes from the facts, from the court's reasoning, and from disposition language that survives masking?

RQ2. Do facts alone retrieve the right statutes and precedents better than controls that exploit citation skew or copy text?

RQ3. Does ontology-directed symbolic structure (claims, elements) preserve outcome signal compared with a data-induced representation, and can it still serve as a checkable explanation layer?

RQ4. Can a signed citation graph tell whether a retrieved authority is still good law?

We do not claim a deployable outcome predictor.

2. Data Sets

Resource	Description
Legal AI Dataset (provided)	26,688 Supreme Court of India judgment PDFs, 1950–2025 (6.8 GB, held locally), with Indian Kanoon citation metadata and a spreadsheet identifying 6,954 land and property disputes: the working corpus, 100% joined to PDFs, median 5,302 words. Only about 32.5% of listed "precedents" are genuine case citations, and only 25.6% of 46,904 case-citation edges point inside the corpus, so recall is always reported against *reachable* authorities with the raw figure beside it.
Derived resources	Win/lose labels (two independent labellers on the land subset; 14,168 usable rules-based labels corpus-wide); 96,623 verbatim-grounded facts; a registry of 42 Acts (58,295 statute citations, 61.4% resolved); two splits that group related judgments and forbid time travel, *forum_heldout* (test n=902) and *temporal_2004_2013* (test n=768); a signed citation graph of 8,467 edges, 62.6% carrying a sign (built earlier, to be restored in WP2).
Ontology and gold set	Provided Land and Property Dispute ontology (15 claims, 66 elements, 6 defence families, 4 remedy tiers); 150 gold cases and 214 element judgements to be annotated in this project. ILDC [4] is an optional external check only, as its release is gated.

3. State-of-the-Art Methods and Algorithms

Outcome prediction. Aletras et al. [1] predicted decisions from text with linear models; Chalkidis et al. [2] and LEGAL-BERT [3] moved to neural encoders. For India, ILDC [4] introduced court judgment prediction and explanation, with the best reported accuracy near 78% on the final 512 tokens. Nigam and Deroy [5] ran facts-only on ILDC and found it worse. Watson et al. [6] show that a model trained on the 4% of features flagged as leakage in UK tribunal claims performs strongly, which general shortcut-learning results [7] predict.

Ontology and extraction. Sadowski and Chudziak [8] measure semantic loss when legal text is ontologised, but do not compare induced and authored vocabularies, the control we run.

Retrieval and graphs. BM25 [9], dense embeddings (Qwen3-Embedding [10]) and reciprocal rank fusion [11] are the standard retrieval stack; HITS [12] and signed variants rank authorities from citation structure; gSpan [13] mines frequent subgraphs. None of the work above verifies every cited authority against real citation data or ranks authorities by signed citation polarity.

4. Research Plan

4.1 Groundwork complete (feasibility evidence)

The pipeline is built and measured on held-out data (48 tests pass); every figure is read from a stored result file.

Stage	Result (held-out)	Control
Precedents from facts	Recall@10 0.238 (BM25 + dense, RRF)	Popularity 0.032 (7.4×)
Statutes from facts	Micro-F1 0.173 on provisions not named in the text	Claim-family prior 0.093 (1.9×); copy control 0.000
Fact families (soft)	AUROC 0.633 from 345 dimensions	0.629 from 47,354 text features
Element layer	0.62% fabricated quotes; 96.9% test–retest agreement; AUROC 0.521	Fails acceptance vs soft facts (−0.11, CI excludes 0)
Leakage (RQ1)	Final-512 input +0.176 AUROC over facts; boundary near 128 words	Volume-matched windows; all 8 decades
Trace faithfulness	Cited facts move the prediction 3.1× as much as random facts	Random and inverse-ranked deletion

4.2 Classical and modern methods, scored identically

Task	Classical / baseline	Modern
Outcome	Per-decade majority; TF-IDF + logistic regression; gradient boosting on six feature groups	Zero-shot, few-shot and chain-of-thought LLMs (llama4-scout-17b, qwen3-235b)
Retrieval	BM25; frequent-itemset and atom-overlap baselines	Dense Qwen3-Embedding-8B; RRF fusion
Extraction	Regular-expression citation and statute rules	Ontology-directed LLM grounding with quote verification
Graph	In-degree, HITS; flat itemsets (498 mined, none significant after Benjamini–Hochberg)	Signed anco-HITS-style ranking; gSpan over fact–element–claim graphs

4.3 Remaining work packages

WP1 Gold evaluation. Annotate the 150 cases and 214 element judgements, with both members annotating 30 shared cases. This gives the first extraction precision, recall and F1, and tests whether element verdicts are *correct*, not merely well-formed.

WP2 Signed authority ranking. Restore the signed graph, rank retrieved precedents by proof-support and edge polarity, and test whether it flags overruled or distinguished authorities better than in-degree. Only 174 of 8,467 edges are negative, so we expect a modest effect and will say so.

WP3 Claim-family patterns. Cluster facts per claim family; mine graph patterns with gSpan.

WP4 Proof graph and verifier. Connect facts, elements, authorities and outcome into one graph with support, refutation and decisive-fact edges, and certify or refuse each authority and

quote.

WP5 Demo. A web dashboard where a user pastes facts and sees certified authorities, element verdicts with quotes, and the trace.

5. Evaluation Plan

Each stage is scored on its own task, win/lose last, and every number needs a control: popularity (precedents), copy-from-text (statutes), per-decade majority (outcomes), volume-matched windows (leakage).

Aspect	Protocol
Splits	*forum_heldout* and *temporal_2004_2013*; every result file records its split hash.
Outcome	Accuracy, macro-F1, AUROC and calibration (ECE) with paired-bootstrap 95% confidence intervals; McNemar tests for paired classifiers.
Retrieval	Recall@k, MRR and nDCG@10 against reachable authorities, with a check that each result pre-dates the query.
Extraction, elements	Per-class precision and recall against gold, Cohen's κ between the two annotators, and the 3.1% self-disagreement of repeated model runs as the floor.
Verification	Share of displayed authorities that resolve to a corpus judgment; share of quotes refused by the quote gate.

Targets. Recall@10 ≥ 0.20; quote fabrication $\leq 1\%$; inter-annotator $\kappa \geq 0.6$; and a reported confidence interval for signed ranking against in-degree, whichever way it falls.

Risks. Outcome AUROC may stay near chance (every confidence interval spans about 0.11), the vocabulary is 20.3% out-of-vocabulary against a 5% freeze threshold, and certification proves provenance, not relevance, so the demo reports both.

6. Project Timeline

Dates (2026)	Task and deliverable
Oct 7–21	Proposal; freeze splits; WP1 gold annotation, extraction P/R/F1, κ
Oct 14–28	WP2 signed graph restored; ranking vs in-degree and HITS
Oct 21–Nov 11	Element correctness against gold; WP3 gSpan patterns per claim family
Oct 28–Nov 18	Class presentation in assigned slot (slides ready by Oct 26)
Nov 4–20	WP4 proof graph, verifier and trace integration
Nov 9–22	WP5 dashboard; end-to-end test on 10 unseen post-2000 cases
Nov 23–Dec 2	Group demo; code and data freeze
Dec 2–9	Final report (15–20 pages) with supplemental code and data, due Dec 9

7. Division of Work

Member	Responsibility
Siddharth Axaykumar Mehta	Pipeline, retrieval, signed graph (WP2), proof graph and verifier (WP4), outcome models and statistics
Om Patel	Gold annotation and element correctness (WP1), patterns (WP3), dashboard (WP5), presentation
Both	Half the gold set each (30 cases shared); weekly meeting; contribution log in the repository; final report

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